# EVERY COHOMOLOGY CLASS ON A RINGED SITE IS KILLED ON A COVERING

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The counted statement and proof contain no cross reference at all, so the assembled document defines every label it uses and no external reference stub is needed.

*Allowed standard prerequisites (editorial).*

- Cohomology of a sheaf on a ringed site via an injective resolution. Standard in this chapter: H^n(U, F) is computed as the cohomology of the complex of sections of an injective resolution, and the restriction map is induced by restricting representatives. Supplies the presentation of H^n(U, F) displayed in the proof and the compatibility of restriction with the representative; the covering argument itself is the counted content.
- An injective resolution is exact in positive degrees. Standard: for n > 0 the sheafified cohomology of the complex I^bullet vanishes, so Im(I^{n-1} -> I^n) = Ker(I^n -> I^{n+1}). Supplies the identification of image and kernel sheaves used to produce the local primitives; locating those primitives on a covering is inside the counted proof.

**Lemma 0.1.** *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $U$ be an object of $\mathcal{C}$. Let $n > 0$ and let $\xi \in H^n(U, \mathcal{F})$. Then there exists a covering $\{U_i \to U\}$ of $\mathcal{C}$ such that $\xi|_{U_i} = 0$ for all $i \in I$.*

*Proof.* Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution. Then

$$H^n(U, \mathcal{F}) = \frac{\mathrm{Ker}(\mathcal{I}^n(U) \to \mathcal{I}^{n+1}(U))}{\mathrm{Im}(\mathcal{I}^{n-1}(U) \to \mathcal{I}^n(U))}.$$

Pick an element $\tilde{\xi} \in \mathcal{I}^n(U)$ representing the cohomology class in the presentation above. Since $\mathcal{I}^\bullet$ is an injective resolution of $\mathcal{F}$ and $n > 0$ we see that the complex $\mathcal{I}^\bullet$ is exact in degree $n$. Hence $\mathrm{Im}(\mathcal{I}^{n-1} \to \mathcal{I}^n) = \mathrm{Ker}(\mathcal{I}^n \to \mathcal{I}^{n+1})$ as sheaves. Since $\tilde{\xi}$ is a section of the kernel sheaf over $U$ we conclude there exists a covering $\{U_i \to U\}$ of the site such that $\tilde{\xi}|_{U_i}$ is the image under $d$ of a section $\xi_i \in \mathcal{I}^{n-1}(U_i)$. By our definition of the restriction $\xi|_{U_i}$ as corresponding to the class of $\tilde{\xi}|_{U_i}$ we conclude. □

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